ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	147302
ToLID	**aRanIta1**
Species	Rana italica
Class	Amphibia
Order	Anura

Genome Traits	Expected	Observed
Haploid size (bp)	5,101,732,283	5,155,001,779
Haploid Number	13 (source: direct)	13
Ploidy	3 (source: ancestor)	2
Sample Sex	unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 6.8.Q64
Obtained EBP quality metric for hap2: 6.8.Q64

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Ploidy is different from Expected

. BUSCO single copy value is less than 90% for hap2

Curator notes

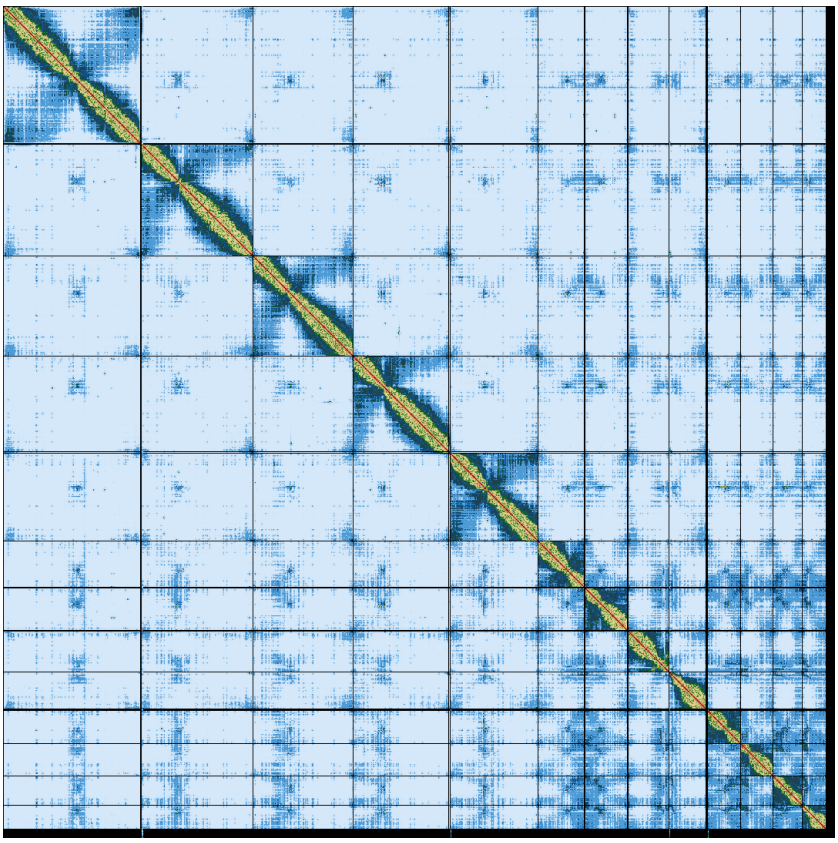
. Interventions/Gb: 33
. Contamination notes: "Both FCS-GX and Blobtoolkit did not detected contaminants in the curated assemblies."
. Other observations: "First, Hifiasm assemblies were scaffolded with yahs without allowing for contig breaks. Then it was curated using the diploid map to produce the curated haplotypes and then reviewed individually with high resolution (HR). In total curation and review, made 26 cuts in contigs, 118 breaks at gaps and 189 joins. Finally, the mitogenome was assembled into a single circular contig 22,497 bp long that will be uploaded to ENA together with the reviewed assembly. The links provided point to a shared folder with the High Resolution maps (HR), for each hap or both (diploid), allowing multimappings (mq0) or not (mq10). Please start your review of from the aRanIta1_*_mq10.extensions.HR.pretext.savestate_1 of each hap.After examining the diploid map and WGAs, for hap2 the SUPER names were switched between SUPER_3-4 and SUPER_10-11 to ensure chromosome correspondence with hap1."

Quality metrics table

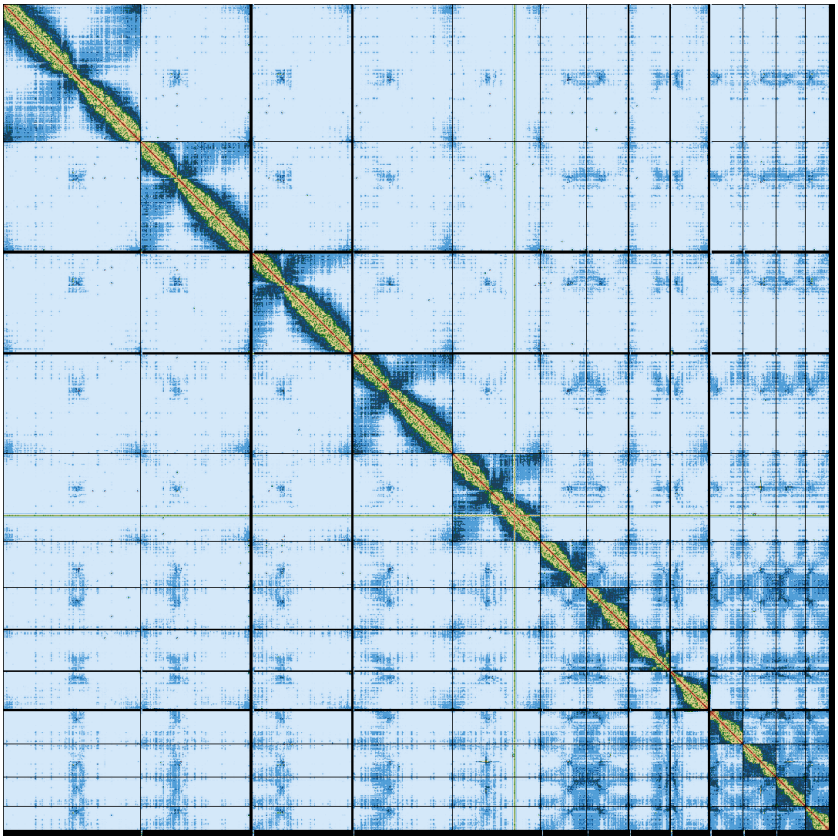
Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	5,149,033,152	5,028,367,072	5,155,001,779	5,021,744,094
GC %	44.56	44.5	44.56	44.5
Gaps/Gbp	502.42	532.78	508.83	540.05
Total gap bp	517,400	535,800	524,600	542,400
Scaffolds	344	330	331	297
Scaffold N50	676,956,069	596,098,675	593,253,258	598,376,166
Scaffold L50	3	4	4	4
Scaffold L90	11	11	11	11
Contigs	2,931	3,009	2,954	3,009
Contig N50	5,393,737	4,359,970	5,334,790	4,359,970
Contig L50	279	317	283	318
Contig L90	1,186	1,351	1,203	1,352
QV	64.094	64.3927	64.0829	64.4007
Kmer compl.	94.8376	94.0275	94.8402	93.9318
BUSCO sing.	90.1%	87.5%	90.0%	87.5%
BUSCO dupl.	2.9%	2.7%	2.9%	2.7%
BUSCO frag.	1.9%	2.3%	2.0%	2.3%
BUSCO miss.	5.1%	7.5%	5.1%	7.6%

BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: vertebrata_odb12 (genomes:119, BUSCOs:3390)

HiC contact map of curated assembly

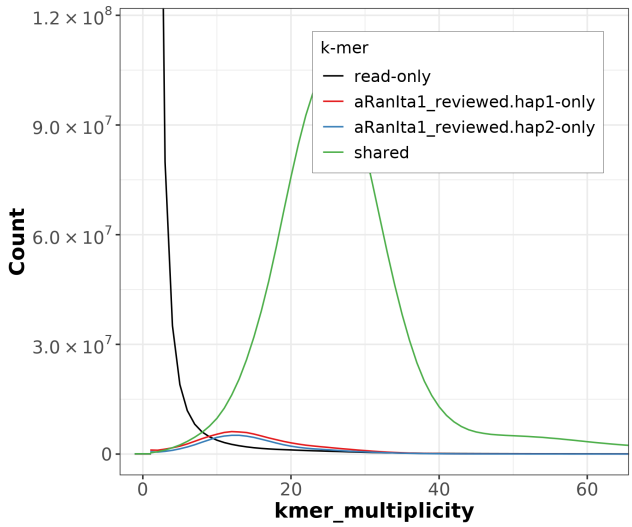


hap1 [LINK]

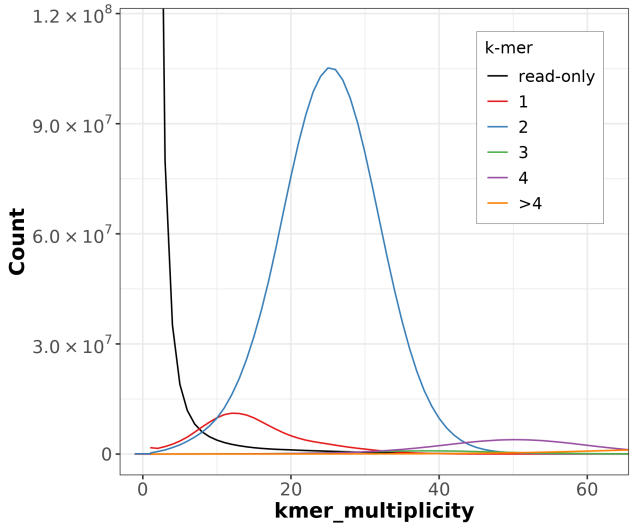


hap2 [LINK]

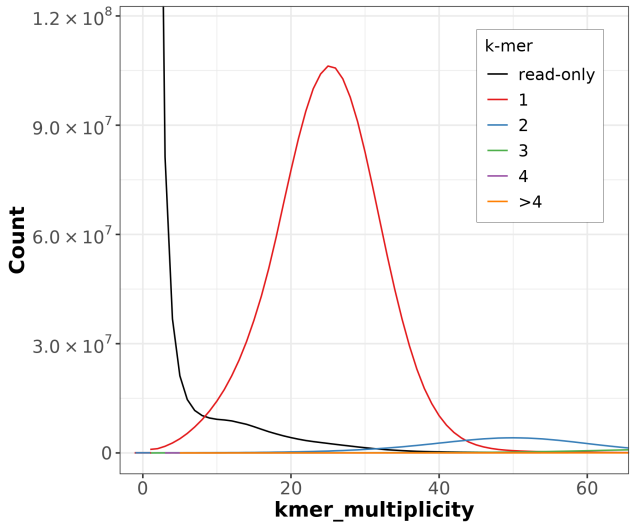
K-mer spectra of curated assembly



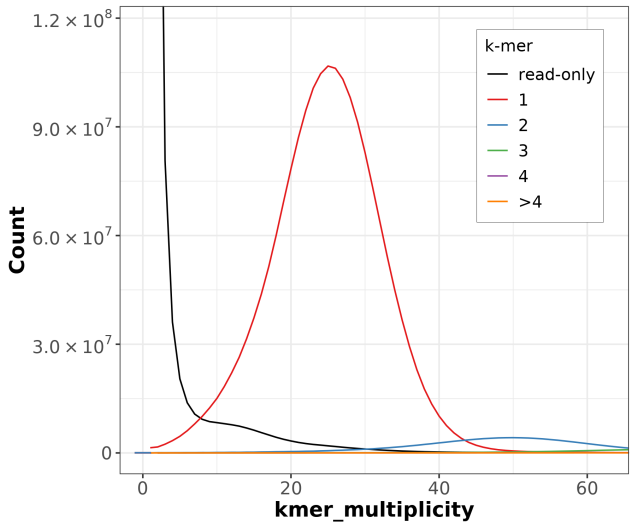
Distribution of k-mer counts coloured by their presence in reads/assemblies



Distribution of k-mer counts per copy numbers found in asm (dipl.)

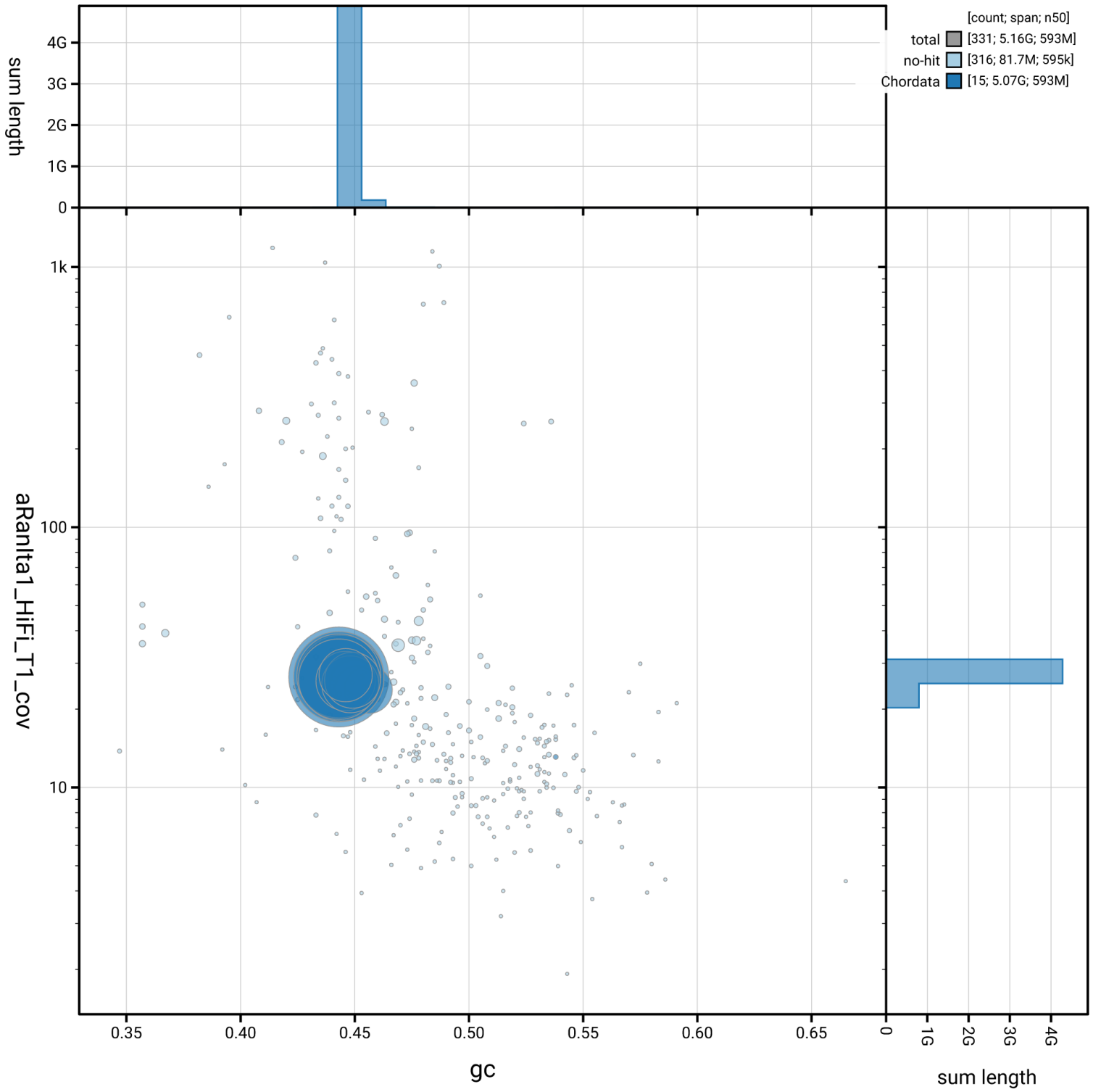


Distribution of k-mer counts per copy numbers found in **hap2** (hapl.)

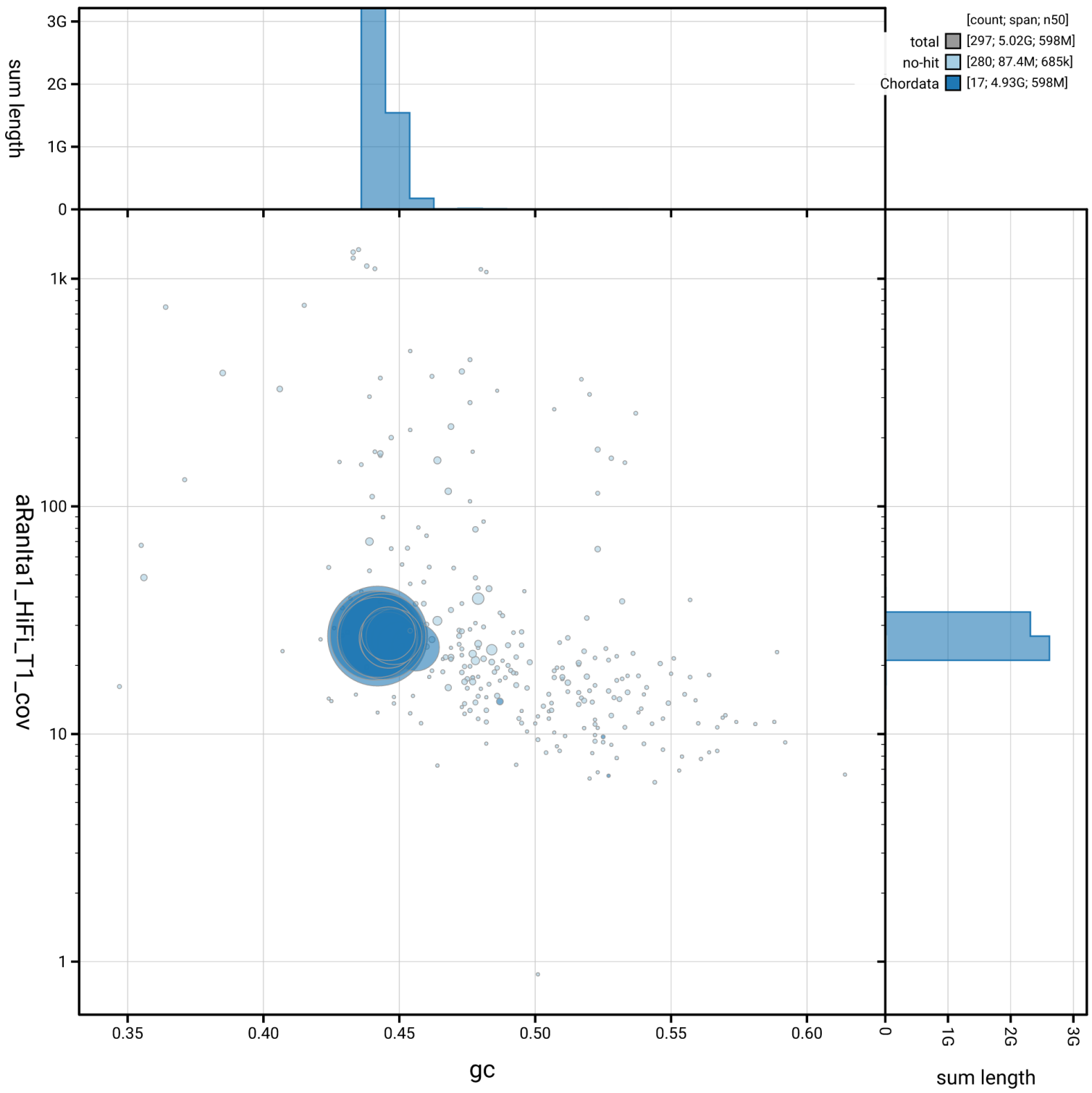


Distribution of k-mer counts per copy numbers found in **hap1** (hapl.)

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.



hap2. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	Hi-C
Coverage	27x	240x

Assembly pipeline

- **CLAWS pipeline**
    |_ *ver:* 3.1
    |_ *key param:* NA
- **Hifiasm**
    |_ *ver:* 0.24.0
    |_ *key param:* --telo-m TTAGGG
- **YaHS**
    |_ *ver:* 1.2a
    |_ *key param:* --no-contig-ec
- **Merqury**
    |_ *ver:* 1.3
    |_ *key param:* NA
- **BUSCO**
    |_ *ver:* 6.0.0
    |_ *key param:* NA
- **tidk**
    |_ *ver:* 0.2.65
    |_ *key param:* NA
- **chromap**
    |_ *ver:* 0.3.2
    |_ *key param:* NA
- **FOAM pipeline**
    |_ *ver:* 0.5
    |_ *key param:* NA
- **mitos**
    |_ *ver:* 2.1.3
    |_ *key param:* NA

Curation pipeline

- **Blobtoolkit DToL Nextflow pipeline**
    |_ *ver:* 0.6.0
    |_ *key param:* NA
- **FCS-GX**
    |_ *ver:* 0.5.5
    |_ *key param:* NA
- **PretextViewAI**
    |_ *ver:* 1.0.5
    |_ *key param:* NA
- **SANGER tola-agp-tpf-utils**
    |_ *ver:* 1.2.1
    |_ *key param:* NA

Submitter: Fernando Cruz

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